Assignment#2

Deadline: 25th oct 2026

Q.1

A bank is redesigning its ATM interface. In the new design, the "Withdraw Cash" and "Deposit Cash" options are placed prominently on the main screen, while less frequently used options such as "Change PIN" and "Print Statement" are placed in a secondary menu.

The designer explains: *"We placed the most frequently used operations on the main screen so that users can access them quickly, while less frequently used operations are kept in a secondary menu to reduce clutter."*

a. What HCI concept is demonstrated by the designer's explanation? Choose from the following:
Frequency of use, Task analysis, Cognitive load, Menu design.

b. What is the design rationale for placing frequently used operations on the main screen?

c. How does this design decision improve the usability of the ATM. Explain using the principles of learnability, flexibility, or robustness?

Q.2

A popular food delivery app like Foodpanda or Uber Eats has been receiving complaints from users. Many customers say they cannot find the "Apply Coupon" button during checkout. The "Place Order" button is too small and often confused with the "Back" button. After placing an order, they are not sure if the order was confirmed. The delivery address form has fields that are difficult to fill on a mobile screen.

a. Open the app and take screenshots of the order flow. Identify four specific usability problems in the ordering process. For each problem, explain which HCI principle is being violated.

b. Find ONE scholarly article about food delivery app usability or mobile checkout design. Summarize ONE key finding from the research. Explain how that finding relates to the problems you identified.

c. Suggest two specific design improvements for each problem. Use concepts like grouping, alignment, affordance, consistency, and feedback. Draw a simple sketch showing your improved design.

Q.3

A large university wants to replace its old student portal with a new system called EduConnect. The system will be used by three main groups: students, teachers, and admin staff. Students will use it to register courses, view grades, pay fees, and check schedules. Teachers will use it to upload marks, share course material, and mark attendance. Admin staff will use it to manage student records, generate reports, and handle admissions. The old portal has many usability problems: students complain about confusing navigation, teachers find it slow to upload marks, and admin staff struggle with generating reports. The university has hired your team as HCI

consultants to redesign the system from scratch. The new system must be usable, efficient, and well-designed for all three user groups. The system must be fully ready before the next semester begins. Your job is to apply HCI principles to improve the design before launch. You will focus on user research, usability engineering, design rules, and navigation design.

a. Explain why skipping user research would be a serious mistake for the EduConnect project. Use the concept of user focus

b. Create TWO personas for EduConnect:

- One for a student (young, busy, uses mobile mostly)
- One for a teacher (senior, less comfortable with technology)

c. Suggest TWO cultural probe methods the university could use to understand its users better. Explain how each method would help.

d. Explain why iterative design and prototyping is important for EduConnect. Give TWO specific examples of how prototyping could help identify problems before launch.

e. Explain the Wizard of Oz technique and why it would be useful for testing the EduConnect mobile app before it is fully built.

f. Apply FOUR of Shneiderman's 8 Golden Rules to EduConnect. For each rule, explain how the design team could implement it and give a specific example.

g. Draw a hierarchical diagram for EduConnect showing the main sections:

- Login
- Student Module (Course Registration, Grades, Fee Payment, Schedule)
- Teacher Module (Upload Marks, Course Material, Attendance)
- Admin Module (Student Records, Reports, Admissions)